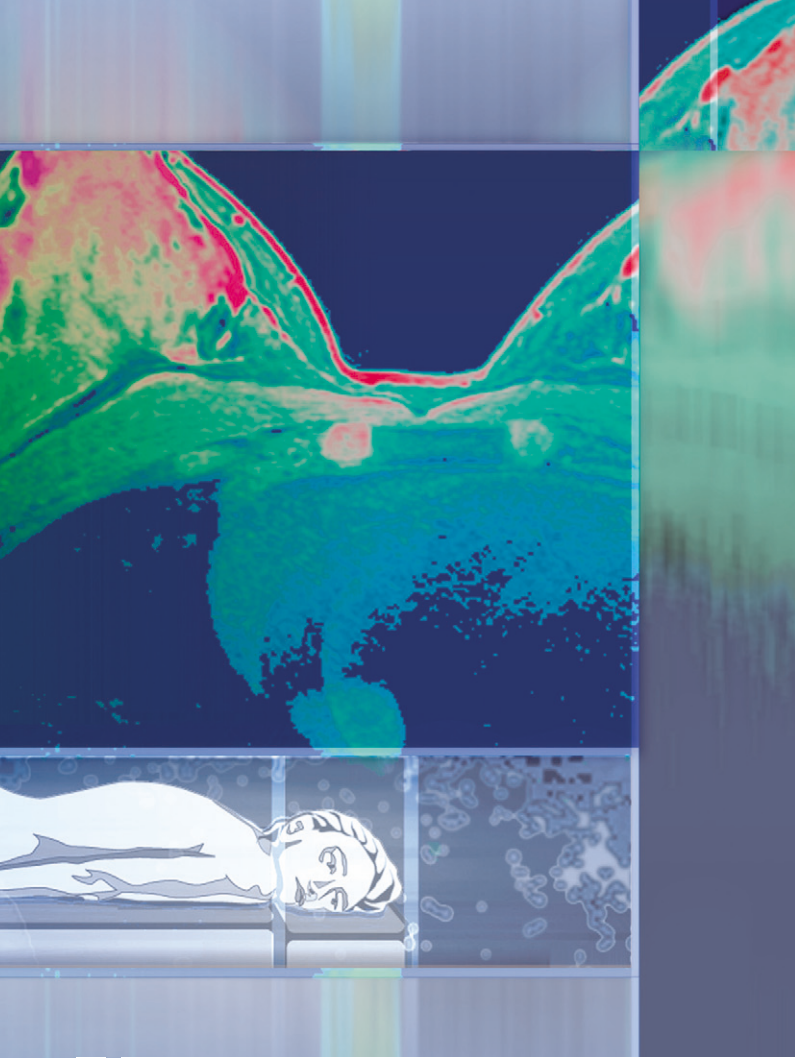

MR Mammography

Application Brochure

MAGNETOM Avanto
MAGNETOM Espree
MAGNETOM Symphony a Tim System
MAGNETOM Trio a Tim System
MAGNETOM Verio

www.siemens.com/healthcare

SIEMENS

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Wittelsbacherplatz 2
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Germany

MR Mammography

Application Brochure

MAGNETOM Avanto
MAGNETOM Espree
MAGNETOM Symphony a Tim System
MAGNETOM Trio a Tim System
MAGNETOM Verio

This brochure describes MR mammography used together with *syngo* MR.
It has been developed for medical personnel working in the field of MR tomography.

To optimize the user-friendliness of this brochure, its contents have been divided into four subject matters.
The first part of the brochure focuses on the basics and provides know-how about the subject matter.
The second part describes practical applications and their use based on sample examinations. The third and fourth part provides the basics as well as use with prosthetic silicone implants.

MAGNETOM Espree – Pink/*syngo* BreVis:
For the MAGNETOM Espree – Pink and the *syngo* BreVis application, several enhanced post-processing tools are available that are described in the BreVis manual.
Furthermore, the *syngo* BreVis software provides its own motion correction, subtraction, and time curve evaluation tools.

Contents

Overview of MR Mammography

MR Mammography is the most sensitive method for detecting and staging invasive breast carcinomas. A special application is used in areas around prosthetic implants.

Suited for

- Verification of invasive carcinoma
- Prosthetic implants (complications, exclusion of tumors)
- Evaluation of benignity and malignancy in cases of structural thickening
- Preoperative determination of multicentricity or multifocality for breast-preserving therapy
- Distinction between mastopathies and malignant tumor formations
- Differentiation of scar tissue
- Follow-up control after tumor surgery

Not suited for

- Evaluation of microcalcification
- Inflammatory changes
- Mammary secretions
- Known mastopathies

Typical breast diagnosis sequence

Medical history and clinical examination

Mammography

Sonography

MR Mammography

Application

MR Mammography is a valuable supplemental method for detecting breast lesions. MR Mammography is the method of choice when diagnosing prosthetic complications.

Sensitivity/Specificity > p. 4

Sensitivity and specificity for breast lesions

To date, there is no general agreement regarding the role of MR imaging in the detection and characterization of breast lesions.

Sensitivity

Probability that the disease exists if the test results are positive.

Author	Sensitivity
Allgayer 1993	90%
Harms 1993	93%
Heywang 1989	100%
Allgayer 1995	96%
Kaiser 1993	99%
Boetes 1994	95%
Fischer 1993	95%
Kuhl 1999	91%
Drew 1999	100%

Extremely high sensitivity is achieved when the carcinoma measures 3 mm or more.

Specificity Probability that the test results will be negative, if the disease does *not* exist.

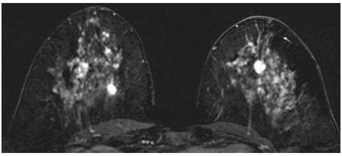
Author	Specif.	Initial	Curve	Morph
Allgayer 1993	29%	–	–	–
Harms 1993	37%	–	–	–
Heywang 1989	42%	+	–	–
Allgayer 1995	51%	–	–	–
Kaiser 1993	97%	+	+	–
Boetes 1994	86%	+	–	+
Fischer 1993	90%	+	+	+
Kuhl 1999	37%	+	–	–
Drew 1993	83%	+	+	–

The specificity of MR Mammography is only moderate compared to the sensitivity. For this reason, limit contrast-enhanced MR imaging to specific indications and use it as an auxiliary method only.

Qualitative evaluations for breast lesions

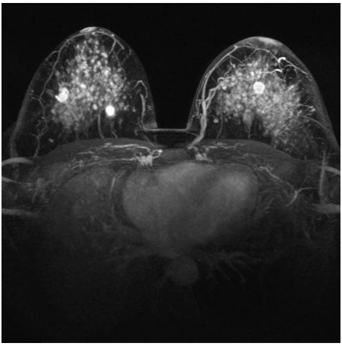
For the detection of breast lesions, MR images as such are *not* informative. An evaluation of the dynamic image data is required to diagnose breast lesions.

Subtraction image



Subtraction
In subtraction images, areas enriched with contrast agent are displayed bright. In this case, a native data set (without contrast agent) is subtracted from a data set after administration of contrast agent.

MIP image



MIP

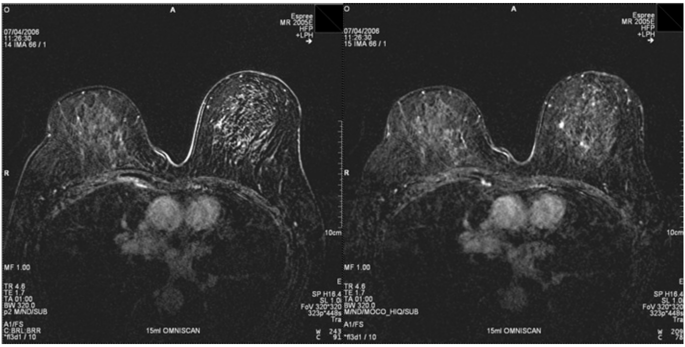
You can also calculate MIP images (MIP = Maximum Intensity Projection) to display vascular courses. The pixels with the maximum intensity are displayed in the MIP image.

Motion correction for dynamic contrast-supported measurements

To diagnose tumors in contrast agent-supported MR images, it has to be ensured that the patient does not move during the entire measurement.
Measurements with contrast agent usually take between 6–8 minutes. This means that motion artifacts may occur. These movements especially complicate diagnoses of very small (multifocal) tumors.

Motion correction

These artifacts can be corrected retroactively with motion correction. This means that even heavily blurred MR measurements can be used for the diagnosis.



Subtraction analysis of a contrast-supported dynamic measurement:
without motion correction (left)
after motion correction (right)

Quantitative evaluations for breast lesions

Certain evaluation of the benignity or malignancy of tumors requires an additional examination of the temporal course of contrast agent enhancement.

Parameter images

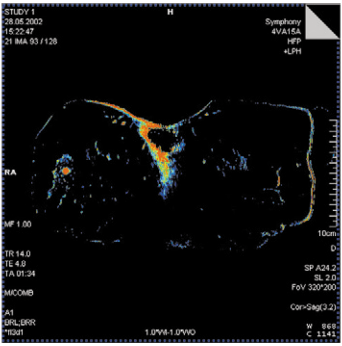
Images that display the temporal course of contrast agent dynamics in color. Standard protocols from Siemens are optimized to spatial and temporal resolution and serve as the basis for parameter image reconstruction. In parameter images, pixels of different tissue types show a different signal behavior.

Parameters to characterize dynamics

- TTP: Time to maximum enrichment (*time to peak*)
- PEI: Area under the enrichment curve (*positive enhancement integral)*
- Washin: Signal increase in the area of enrichment
- Washout: Signal behavior after enrichment
- MIPt: Greatest signal increase during the dynamic measurement

Combination images

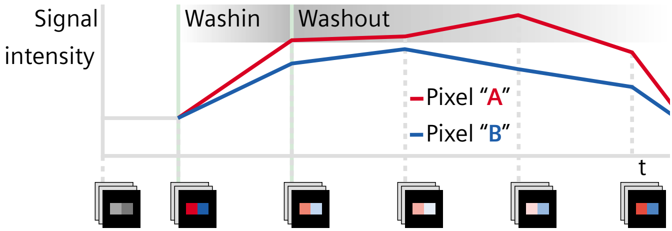
Images that simultaneously display multiple parameters, typically Washin combined with Washout.



Washin/Washout combination image

Parameter images (Washin/Washout)

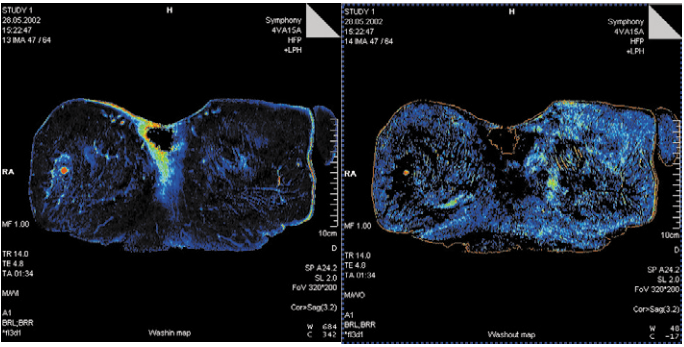
The Washin/Washout parameter images display signal changes at the starting and end areas of the dynamic curves. After contrast agent injection, breast carcinomas display a typical increase in signal intensity with earlier plateau formation and a wash-out effect.



Pixel-by-pixel, the percentage change in signal value per minute is calculated based on the initial series, and is converted into gray and color values.

Highest value During the washin calculation, the parameter "Highest value" ensures that the signal change is measured up to the maximum. This prevents "false" results in case there is a series between the start and end series with a higher signal intensity than the end series.

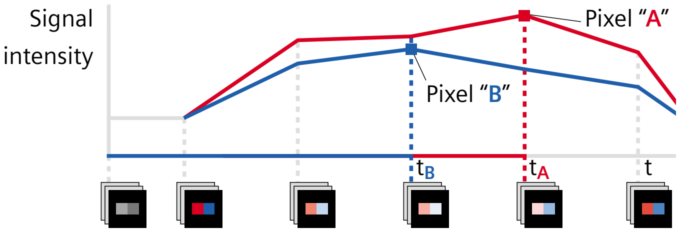
Down sloping washout curve. Strongly negative values in the card are suspect. For this reason, a color palette with an inverted course of colors is used for the evaluation.



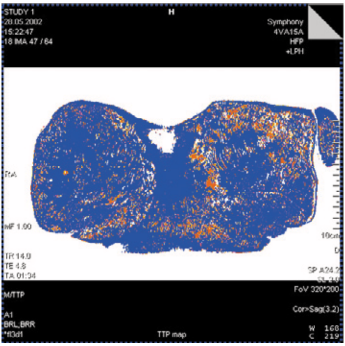
Washin image (left): The rise in signal in the starting area of the dynamic curve is displayed.
Washout image (right): The signal behavior at the end area of the dynamic curve is displayed.

Parameter images (TTP, PEI)

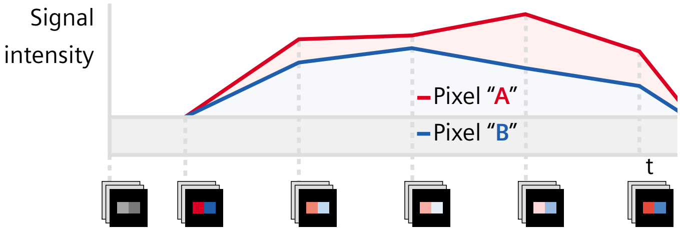
The time-to-peak image displays (pixel-by-pixel) the time in seconds until signal maximum. In a typical dynamic MR breast examination, multiple identical measurements (e.g., 6–8) are acquired, yielding just as many time points (6–8) for the dynamic curve evaluation and color maps.



TTP image

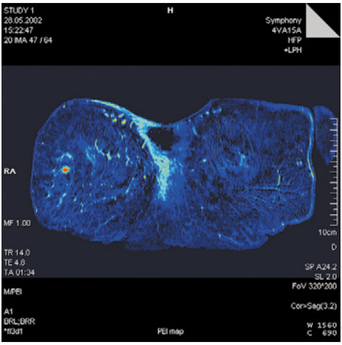


The positive enhancement integral image displays (pixel-by-pixel) the area under the time intensity curve.



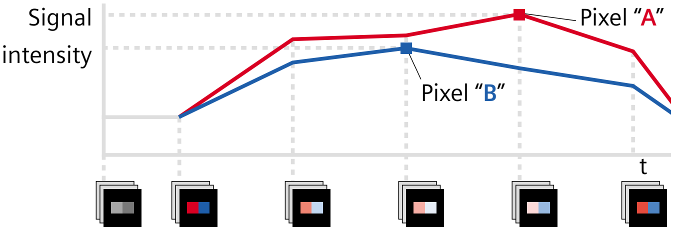
Positive enhancement integral for two different pixels. The value in the results image is greater for Pixel "A" than for Pixel "B".

PEI image



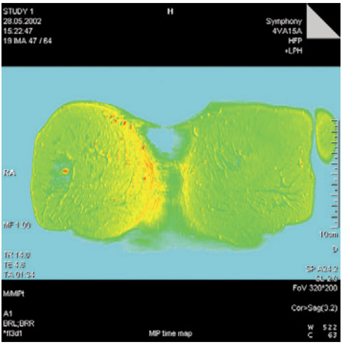
Parameter images (MIPt)

The MIPt calculates a Maximum Intensity Projection over time for each pixel. In the MIPt-image, you can see the extent of the increase in signal.



The brightest pixel in the dynamic time range is used for the results image.

MIPt image



Dynamic examination overview

Precise detection of breast lesions is the goal.
For this reason, we measure T1-weighted precontrast and postcontrast images and then perform an evaluation (subtraction, parameter images, etc.).

Preparation

- **Prepare patient**
- **Prepare contrast agent injection**

Measurement (inline)

- **Diffusion-weighted measurements with REVEAL (optional)**
- **Set dynamic measurement**
- **Set subtraction**
- **Select parameter images**
- **Perform dynamic measurement**

PROTOCOL
Localizer
3D FLASH dynamic
VIEWS 1+5M
(1 precontrast, 5 postcontrast measurements, incl. inline reconstruction of parameter images)

Post-processing

Perform motion correction (if required)

Reconstruct combination images

Mean Curve

Manual post-processing

Manually scale subtraction images

You can also use transverse slices with the GRAPPA iPAT option (PAT factor 2). This ensures the same temporal resolution as with coronal slices. Additionally, transverse slices enable good evaluation of the ductal course.

Prepare contrast agent injection

Position the patient prone in the breast coil.

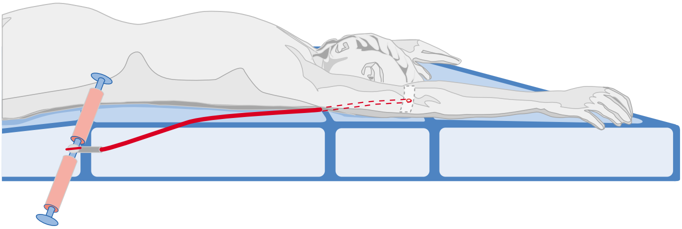
Recommendations. For MAGNETOM Espree and Verio: Position the breast coil on the spine matrix so that the breast is closer to the isocenter of the magnet.

Route the tube for the infusion *prior to* moving the patient table into the magnet.

Recommendations. The tube should be long enough so that it can be accessed from outside when the patient is in the magnet bore.

Insert a cannula into the forearm vein. Connect it to the injection tube with the attached 3 -way stop valve.
Fill the contrast agent and saline solution in two separate syringes and connect the syringes to the stop valve outside the magnet bore.

Recommendations. Do *not* use I.V. bottles.



Instruct patient. Prior to moving the patient into the magnet, explain that she should remain completely still during the entire examination.

Diffusion-weighted measurements with REVEAL (optional)

For additional evaluation of lesions or follow-up, you are able to measure diffusion-weighted images with REVEAL. For this purpose, the lesion made visible in the dynamic examination has to be located in the diffusion-weighted images.

Recommendations. Since the contrast agent affects the T1 relaxation time and thus the effectivity of the fat suppression, an EPI-STIR measurement should be taken *before* the dynamic measurement.

REVEAL protocols. In the protocol tree under breast/library/other; with inline-ADC computation (prerequisite: Inline Diffusion option).

Determining the ADC value
The ADC map (ADC = Apparent Diffusion Coefficient) allows for an in-vivo pseudo quantification of the diffusion effect. Due to the limited mobility of water molecules in the vicinity of a tumor, malignant lesions show lower ADC values than benign lesions.

Measurement (inline)

DWI with REVEAL

Transfer an ROI within the lesion from the diffusion-weighted images to the ADC map.

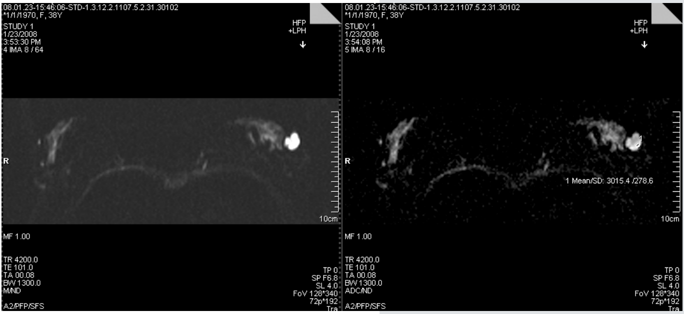
For this purpose, go to the Viewing task card and draw an ROI into the diffusion-weighted image.

Copy the ROI to the ADC map.

Determine the ADC value (mean value of the grey-scale values).

TIP

Drawing ROIs: Central necroses should not be enclosed by an ROI. It has a different cellularity and therefore a different ADC value compared to the tumor.

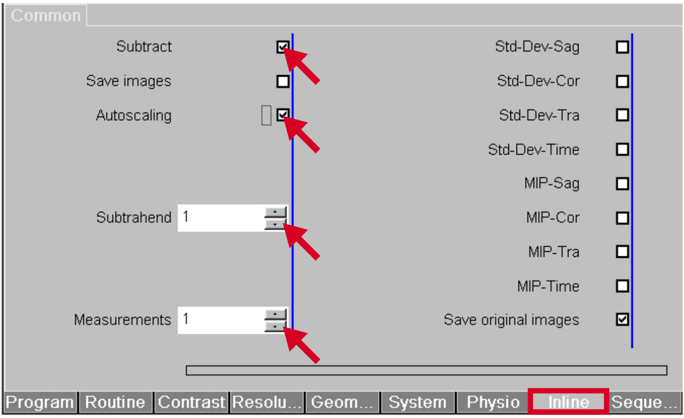


Set dynamic measurement (inline)

Localizer is measured/ dynamic protocol is open

Set any additional parameters desired in the dynamic protocol.

Please note: Use multiple measurements to prevent repeated adjustments between measurements.



Inline card

Measurement (inline)

Set dynamic measurement

Set subtraction

Parameters for subtraction
The parameters are optimized in the Siemens standard protocol and usually may not be changed.

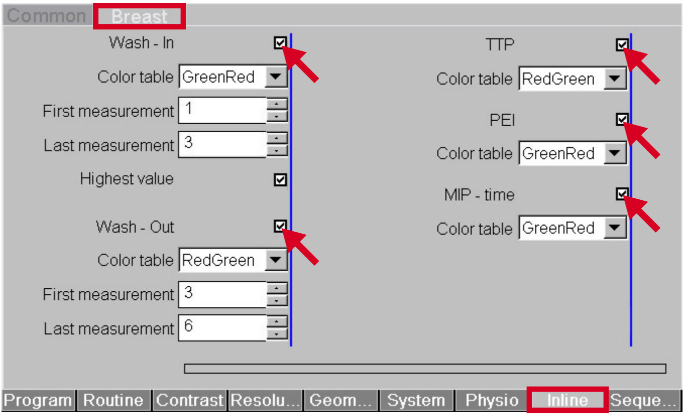
The number of the last measurement without contrast agent is set as the "Subtrahend"
"Autoscaling" ensures that any negative values that occur during the subtraction are *not* set to zero.

TIP
In addition, you can create MIP images in reconstruction directions sag, cor, and tra.

Manually scale subtraction images > p. 38

Select parameter images (inline)

If you would like to activate inline computation for dynamic protocols, please use the Inline Breast card.



Inline Breast card

Select the parameter images desired using the Inline Breast parameter card.

Measurement (inline)

Select parameter images

Reconstruct Washin image
Set the start value to the last precontrast series and the end value to the series in which the enrichment phase ended.

Reconstruct Washout image
Set the end value from the Washin parameter as the start value, and the last series as the end value.

TIP
You can also have the parameter images computed retroactively.

TIP
Windowing Washout images: Orientation to the contrast agent washout at the heart can be helpful.

TIP
Windowing can be made easier by using the **Perfusion** color table.

Perform dynamic measurement (inline)

Parameters for the dynamic measurement have been set

Check the infusion site and patency of the infusion tube immediately prior to starting the dynamic series.

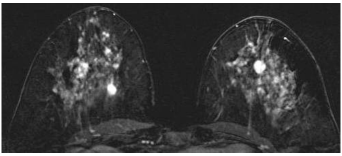
Start the dynamic precontrast series.

Inject the contrast agent bolus immediately after the first measurement and then irrigate with a sufficient quantity of saline solution.
The subsequent postcontrast measurements start automatically.

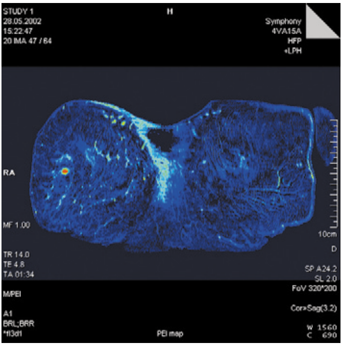
You obtain dynamic precontrast and postcontrast series as well as subtraction and parameter images.

Measurement (inline)

Perform dynamic measurement



Subtraction image



PEI parameter image

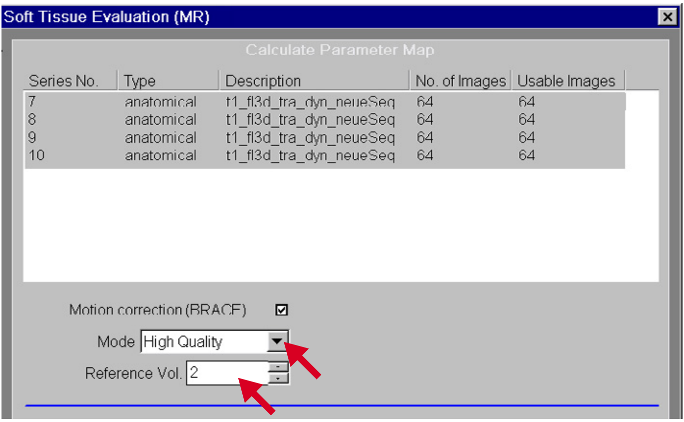
Performing motion correction (if required)

Measurement was performed

MR images that are either not suitable for a diagnosis or only to a limited degree due to motion, can be corrected retroactively.

Two different motion correction algorithms are available for this purpose:

- "Fast" mode
  Suitable for small to medium strong motion artifacts, faster computation times.
- "High Quality" mode
  For very strong motion artifacts, however, with longer computation times.



Post-processing

Perform motion correction

Select the desired motion correction mode.

Determine which measurement you want to use as a reference measurement for motion correction. (Reference volume)

TIP

The reference measurement should be the measurement where highest signal intensity can be expected in strongly enhanced tumors. This is the first post-contrast measurement for dynamic measurements with a temporal resolution between 60 to 90 seconds. For this reason select No. 2 as the reference volume.

Reconstructing combination images (offline)

Parameter images are available

Optimum selection of all parameters is required to obtain a meaningful combination image. For this reason, only subsequent reconstruction of combination images is available.

Select e.g., a Washin and Washout series from the Patient Browser.
In the menu, select “Applications > Breast (MR)”.

Use the weighting factor to determine the extent to which each of the parameter images should contribute to the result.
The default value for Washout is – 1.0 and for Washin it is + 1.0.

Effect of the weighting factor

Weighting factor	Parameter image	Combination image
negative	small pixel value	bright pixel
	large pixel value	dark pixel
positive	small pixel value	dark pixel
	large pixel value	bright pixel

Post-processing

Reconstruct combination images

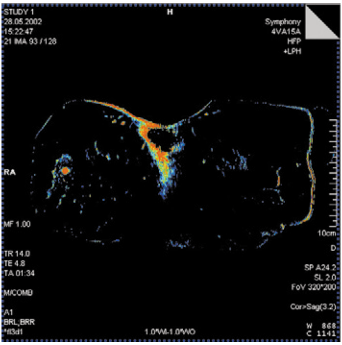
Use the "Window" and "Center" parameters to set how the brightness and contrast of the parameter images will contribute to the combination image.

Please note: The following rule applies when using fixed values.

Pixel value ≤ Window – Center / 2 = lowest gray scale or color value
Pixel value ≥ Window + Center / 2 = largest gray scale or color value

Start the reconstruction of combination images with "Evaluate".

TIP
Take the optimum setting for brightness and contrast from the displayed windowing values in the Viewer (W/C) or set fixed values.



Washin/ Washout combination image

Evaluate contrast agent dynamics (offline)

Precontrast/ postcontrast images are available

Use the Mean Curve to evaluate the contrast agent dynamics.

Select all precontrast and postcontrast series in the Patient Browser.

In the menu select "Applications > Mean Curve".
The images are loaded in the upper left segment.

In the Patient Browser, select the subtraction series in which you want to draw the ROI, and use drag & drop to move the image to the lower left segment.
For improved localization of suspicious areas, you can also load an additional reference image (e.g., sagittal MIP image or parameter image) in the lower right segment.

Select the subtraction image you want to evaluate.
Copy this slice position to the upper left segment.

Post-processing

Mean Curve

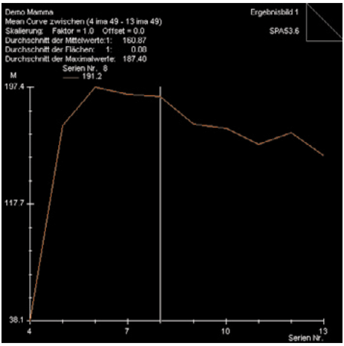
Draw the desired ROIs into the subtraction image.

The drawn ROIs are automatically transferred into the upper left segment.

Start the evaluation.

Confirm with “OK” that the pre-contrast series was taken into account as a reference series for the evaluation.

For results, you receive a curve evaluation and, behind it, a tabular evaluation, in the upper right segment.



Curve evaluation

TIP

Storing the evaluation, including the anatomical image with the ROI:

Manual post-processing (offline)

If you have not selected inline reconstruction for the dynamic measurement, upon completion of the measurement you can reconstruct subtraction images from the series obtained and statistically analyze parameter images.

Manual subtraction
Select the precontrast and one or more postcontrast series in the Patient Browser.

In the menu, select “Evaluation > Dynamic Analysis > Subtraction”.
The precontrast series is automatically subtracted from the postcontrast series.

Click Test to compute an example subtraction image using the parameters currently set.

Click OK to generate the entire subtraction series.
The subtraction series is automatically stored in the Patient Browser.

Manually scale subtraction images > p. 38

Statistically analyze Washin and Washout images

In the Viewer, draw a ROI in the Washin or Washout image.
The minimum and maximum value in the ROI is displayed.

Divide the displayed values by 10. You obtain the minimum and maximum signal increase in the ROI in percent per minute.

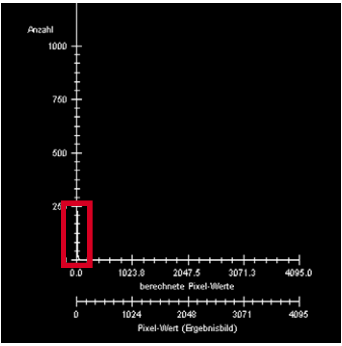
TIP

If the minimum and maximum values are not displayed in the ROI, click in the ROI and open the context menu with the right mouse button.
Select "Minimum" and "Maximum" under menu item "Options".

Manually scale subtraction images (offline)

Adaptation of the intensity range necessary

In some cases during subtraction, the result image may display "black" areas. This means the postcontrast data set has a lower signal intensity than the native data set. In this case, you can scale the intensity range for the subtraction.

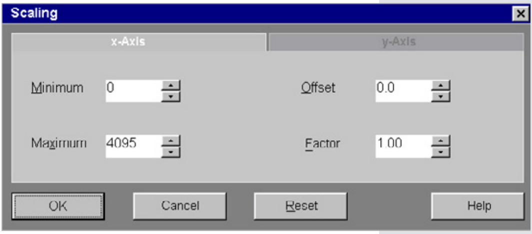


Scaling required:
The intensity values of the pixel are accumulated along the Y axis; i.e., they cannot be displayed in the intensity range.

Post-processing

Manually scale subtraction images

Change the intensity range in the “Scaling” dialog box.
In this case, click the right mouse button in the histogram and then with the left mouse button click “Properties”.



Scaling dialog box

Move the “X-Axis” function card to the foreground.

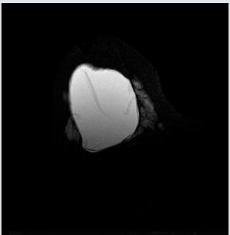
Reset scaling,
e.g., Minimum –64,
Maximum 4032.
The histogram is adjusted accordingly.

Prosthetic silicone implants

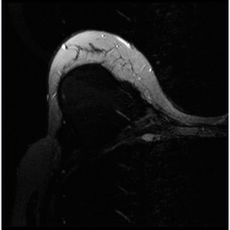
MR imaging of prosthetic silicone implants require images with high resolution. Depending on the clinical question, either fat or silicon can be suppressed, or pure silicon images can be created.

Measurement types

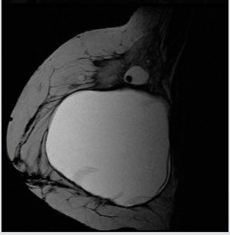
- Pure silicon images: to display capsule contractures, prosthesis dislocation, ruptures (intracapsular: "linguini" signs, extracapsular: defect in the fibrous capsule, possibly siliconoma).
- Silicone suppression: to display fat or tumors.
- Water suppression: to suppress the signal from cysts (water is displayed dark)
- Fat suppression: to display water-filled cysts (water is displayed bright).



pure silicone image



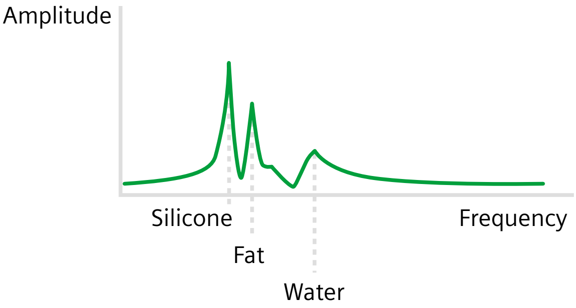
with silicone suppression



with water presaturation

Spectral differences

Spectral differences between fat, silicone, and normal breast parenchyma enable selective signal suppression.



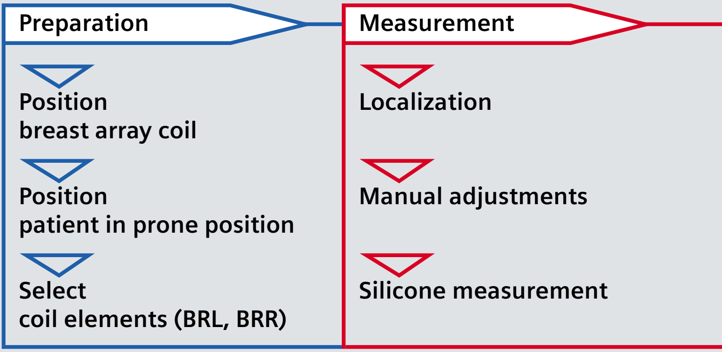
Display of MR spectra: The frequency of spectra (silicone, fat, and water) runs from left to right. Normally, the fat line somewhat overlays the silicone line.

Fat/silicone suppression
Fat or silicone suppression may be complete or partial.

Better orientation. Contours of the breast and implant may be better viewed using partial fat/silicone suppression.

Displaying pure silicone images

It may be necessary to display silicone implants during breast examinations. In this case, the signals from fat and water have to be suppressed. The result image displays the pure silicone signal.



Recommendation. To obtain better adjustment results, we recommend these measurements be performed selectively on one breast only. As indicated under “Preparation”, select coil element BRL or BRR an (depending on the breast in question).

Possible measurements

- STIR with water saturation: the fat signal is suppressed by an additional inversion pulse. The water saturation pulse (centered on the water peak) suppresses the signal from the breast and blood vessels. In the result image, silicone is displayed bright.
- T2 sagittal: to check the impermeability of the silicone implant. Displays creases in the implant (e.g., linguini or keyhole signs)
- Silicone suppression: with any T1-weighted measurement or T2-weighted measurement using a water saturation pulse (centered on the silicone peak). In the result image, silicone is displayed dark.

Spectral water saturation is possible because the frequency separation between water and silicone is greater than that between water and fat. However, silicone might become partially saturated.

Spectral differences > p. 42

Perform manual adjustments (I)

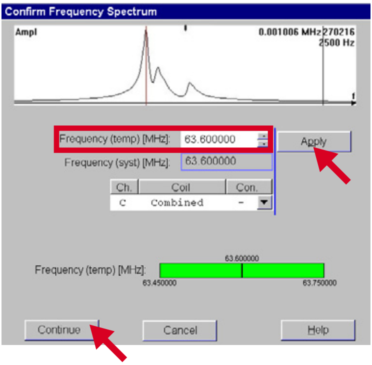
Protocol is selected

Position the slices for the respective protocol (fat saturation, water suppression, etc.)

On the System Adjustments parameter card, select "Confirm freq. adjustment".

Start the measurement.
Adjustment is performed automatically. The "Confirm Frequency Spectrum" dialog window is displayed.

Peaks from left to right: silicone—fat—water



Click "Apply" and "Continue", to copy the desired frequency.

If you want to change the frequency:

Enlarge the frequency spectrum (double-click with the *right* mouse button).

Improving fat saturation
Center the frequency on the fat peak (click the left mouse button on the fat peak).
This field is then copied to the field "Frequency (temp)".

Add 220 Hz at 1.5 Tesla to this value or 440 Hz at 3.0 Tesla (the last 3 numbers of this frequency).

Click "Apply" and "Continue".
The measurement is performed without additional adjustments.

TIP
With very high fat proportions: Check the automatic peak detection prior to each measurement with spectral fat sat and adjust manually, if necessary.

Perform manual adjustments (II)

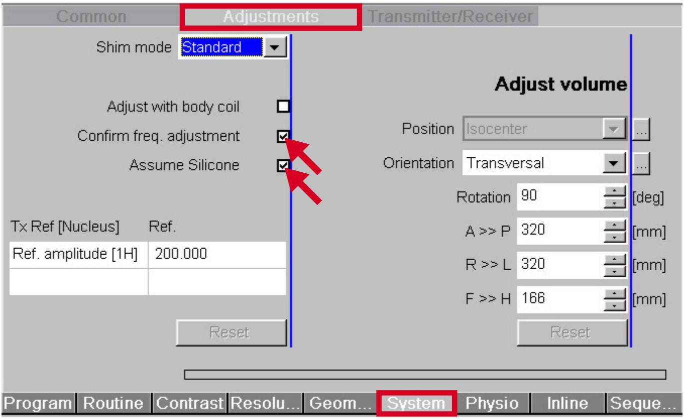
Protocol is selected

Change water saturation
The water saturation suppresses the peak used for centering, and may be utilized for almost every protocol.

On the contrast card of the selected protocol, select "Water Suppression".

Select the system card for the protocol.

Select parameters "Confirm freq. adjustment" and "Assume Silicone".



As soon as the “Confirm Frequency Spectrum” dialog window is displayed:

Click “Apply” and “Continue”, to copy the desired frequency.

If you want to change the frequency:

Click the left mouse button on the frequency peak you want to suppress.

TIP

For unilateral or small implants: Check the automatic peak detection, because the silicone signal can be very small. For this purpose, view the signals of the individual coil elements so you can check whether the system frequency on the side of the implant is located on the desired peak (e.g., water, if water is to be saturated to obtain a silicone image).

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